Presentation 4

1. Link to presentation and code on GitHub:
   - Presentation Link: Click Here (Note: static visuals used in presentation; to see live dashboard use link below) / Click Here (video link)
   - Dashboard Link: Click Here
   - Data cleaning code: Click Here

2. What did you do?

This week, my goal was to build off my previous visualization—which tracks call volumes by month for each menu type—by adding in the data for the calls beyond the caller-facing menus (i.e. Voicemail, ClosedQueue, etc.). This way, Cynthia can easily compare how many callers are there for each menu and submenu type across the relevant time frames that she needs. This visual is supposed to compliment Julia's visual as they seek to display similar information across different time frames; Julia displays each menu's usage over the course of a day whereas mine shows it over the course of a year.

To do so, I started with the combined_calls.csv data set from last week. It is a combination of the weekly CAR datasets between April 2024 and September 2025. Since the data was pretty raw, I had to do some data cleaning in order to get it ready for PowerBI.

` I started off by converting the timestamps in the dataset into standardized date/time components (Year, Month, Day). I handled malformed dates using errors = "coerce." This ensured all timestamps are in a consistent format for time-based analysis and aggregation. After this, I defined the activity mapping by starting with listing out each activity into service menus that would be used for pattern matching. The list included "Farmworker," "Seniors," "HIVMenu," "Family," "Housing," Benefits," "Consumer," "Criminal Records," Employment," "Immigration," "ADAPT," and "Other." This will be helpful for tracking the call volume by service areas in order to understand the demand for different programs. After that, I laid out the session-level call outcome classification that would be used to group all activities within each call session (identified by Contact Session ID) and determine the outcome. The list created four categories: agent answered (call was handled by a staff member), voicemail (call went to voicemail), ClosedQueue (call came in when the queue was closed such as after hours or when the capacity is full), and other (unclassified outcomes). Finally, I counted the number of unique call sessions per day, broke down counts by menu category, subtype (outcome), year, month, and day. I used .nunique() to avoid double-counting sessions that have multiple activity records. Lastly, I rolled up daily data into monthly summaries with two key metrics: 1) unique_sessions_per_month (total unique calls that month) and 2) average_daily_unique_calls (mean calls per day during that month).

In PowerBI, I made two graphics. The first graphic I made is a bar chart that tracks the monthly call volumes between April 2024 and September 2025 for each menu type. On the x-axis, it displays the months of the year and on the y-axis it displays the total call volume for each month. There is a filter to select which menu category and subtype (outcome) you want to look at as well as a hierarchical filter for the year and month you want to hone in on. The last graphic is a bar chart that tracks the average number of daily calls for each month between April 2024 and September 2025 for each menu type. On the x-axis, it displays the months of the year and on the y-axis it displays the average daily call volume for each month. There is a filter to select which menu category and subtype (outcome) you want to look at as well as a hierarchical filter for the year and month you want to hone in on. In the end, these two graphics say the same thing: Most menu lines are experiencing capacity constraints, with Benefits, Employment, and Family showing the highest ClosedQueue rates (50-60%), while only ADAPT and Consumer maintain strong agent answer rates above 60%. Call routing patterns suggest that Farmworker and HIV menus may have navigation issues, as 80-90% and 30-40% of their callers respectively move to other options, and nearly half of all calls never reach an issue-specific menu at all.

3. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it appears most menu lines are experiencing capacity constraints, with Benefits, Employment, and Family showing the highest ClosedQueue rates (50-60%), while only ADAPT and Consumer maintain strong agent answer rates above 60%. Call routing patterns suggest that Farmworker and HIV menus may have navigation issues, as 80-90% and 30-40% of their callers respectively move to other options, and nearly half of all calls never reach an issue-specific menu at all.

With this in mind, we recommend that Legal Aid do three things. First, they should increase capacity during predictable seasonal peaks for menus where ClosedQueue consistently tracks seasonal trends (Benefits, Employment, Housing, Seniors). Across these lines, 40–60% of callers are blocked, and the ClosedQueue rate rises and falls in lockstep with overall call volume over time. This indicates a structural capacity shortfall, not random overload—making it clear that agent staffing or queue hours need to expand during known high-demand periods. Second, LegalAid should redesign routing for menus where long-term trends show rising "Other" volume despite falling total calls (Immigration, HIV, Farmworker). Time trends show that for these menus, the share of callers passing through without receiving help increases over time, even as total call volume declines. This pattern signals growing confusion or misrouting over time, suggesting the need for IVR pathway redesign, clearer naming, or different placement in the menu. Lastly, they should preserve successful operational patterns in menus where positive trends persist (ADAPT, Consumer), but smooth out irregular spikes. ADAPT and Consumer show stable long-term trends in agent pickup (60–75%), and Consumer's ClosedQueue rate declines steadily over the year, suggesting strong processes. ADAPT, however, shows irregular volatility in ClosedQueue spikes over time, recommending finer-grained forecasting and targeted coverage to minimize sudden service interruptions.

4. Issues faced, attempt to resolve issues, and resolved issues:

This week I struggled with the abandonment tagging and the senior submenu type grouping (i.e. SeniorADAPT, SeniorHousing, etc.). While I tried multiple times to resolve this, leveraging resources such as AI and other group mates, I decided to push this to next week.

5. Next Steps

- Add hierarchical filtering for Senior sub-menu options (i.e. Senior ADAPT, SeniorHousing, etc.) in conjunction with the current submenu types (i.e. Agent-Answered, Voicemail, etc.)
- Add in abandonment tagging to track the number of abandoned calls for each menu type over time

- Work closely with teammates analyzing other metrics to link time-trend findings with their insights for a more complete understanding

6. References

- Input code from Krish
- Julia’s data transformations in python
- Information from Cynthia during class with her

Presentation 3

7. Link to presentation and code on GitHub:
    - Presentation Link: Click Here (Note: static visuals used in presentation; to see live dashboard use link below) / Click Here (video link)
        - Note: in the video, the presentation formatting is slightly off as Zoom wouldn't allow us to present google slides so we had to download it as a Keynote file (which messed with the formatting)
    - Dashboard Link: Click Here
    - Data cleaning code: Click Here

8. What did you do?

This week, my goal was to build off my previous visualization—which tracks call volumes by month—by allowing the viewer to see call volumes for each menu while adding hierarchical filtering by year and month. This way, Cynthia can easily compare how many callers are there for each menu across the relevant time frames that she needs. This visual is supposed to compliment Julia's visual as they seek to display similar information across different time frames; Julia displays each menu's usage over the course of a day whereas mine shows it over the course of a year.

To do so, I started with the combined_calls.csv data set from last week. It is a combination of the weekly CAR datasets between April 2024 and September 2025. Since the data was pretty raw, I had to do some data cleaning in order to get it ready for PowerBI.

` I started off by defining a list of target menu names (related to legal issues) that is derived from the activity names column. This list included "Farmworker", "Seniors", "HIVMenu", "Family", "Housing", "Benefits", "Consumer", "Criminal Records", "Employment", "Immigration", and "ADAPT." I only put the names of the issues, rather than the activity name itself, so as to make sure I didn't miss anything due to syntax or slight naming differences. After this, I filtered out rows that contained some version of the above list in the activity name column. Then, I removed any system-level or non-caller facing activities such as "GetLoggedIn", "Play", "Read", "Set", "Callback", "Disconnect", and "Queue." This way we could hone on solely on entries related to specific interactions/decisions made by the caller. After that, I converted timestamps into date components (year, month, day) so calls can be grouped by time. Finally, I grouped the data by activity name and date to calculate how many unique call session IDs occurred for each menu on each day. I also aggregated the daily results by month to find both the total number of unique sessions per month and the average number of daily unique calls for each menu. Lastly, I compared the number of unique call session IDs between the raw dataset and the cleaned data set, revealing that I had removed about 50,000 entries. This can largely be attributed to the removal of rows with non-caller facing activity names (i.e. Disconnect, Queue, etc.) in order to hone in on where callers are interacting with the menu.

In PowerBI, I made three graphics. The first simply tracks the overall monthly call volumes between April 2024 and September 2025 as a line chart. On the x-axis, it displays the months of the year and on the y-axis it displays the total call volume for each month. I made this graphic to lay out the context for my analysis. It showed that in general, Legal Aid is receiving more calls per month in 2025 than in 2024 and that this trend is likely to continue. The second graphic I made is a bar chart that tracks the monthly call volumes between April 2024 and September 2025 for each menu type. On the x-axis, it displays the months of the year and on the y-axis it displays the total call volume for each month. There is a filter to select which menu option you want to look at as well as a hierarchical filter for the year and month you want to hone in on. The last graphic is a bar chart that tracks the average number of daily calls for each month between April 2024 and September 2025 for each menu type. On the x-axis, it displays the months of the year and on the y-axis it displays the average daily call volume for each month. There is a filter to select which menu option you want to look at as well as a hierarchical filter for the year and month you want to hone in on. In the end, these two graphics say the same thing: despite call volumes overall increasing, the Benefits, HIV, Immigration, and SeniorsADAPT menus are seeing decreases in their usage in 2025 compared to 2024.

9. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it appears the Benefits, HIV, Immigration, and SeniorsADAPT menus are seeing decreases in their usage in 2025 compared to 2024. This can largely be attributed to national policy changes such as increased deportations and cuts to funding for programs related to HIV, economic welfare, and elderly care. This implies that despite the number of calls for these menus/issues decreasing, people are likely not facing less problems related to these issues. Rather they feel less comfortable talking about them or do not think they can get help given our political environment.

With this in mind, we recommend that Legal Aid do two things. First, they should create a plain-language online resource library with short, multilingual guides on common legal topics (e.g., benefits appeals, senior rights, immigration updates, HIV-related discrimination) to help build trust, help people self-identify their legal issues, and encourage informed engagement with legal aid services. Second, they should pilot an anonymous chat or text support option to let clients ask initial questions confidentially and safely—especially valuable for sensitive issues like immigration status, health, or benefits—lowering fear and re-engaging communities hesitant to call.

10. Issues faced, attempt to resolve issues, and resolved issues:

I didn't face many significant issues during this revision. In the past weeks, I think Julia and I were not collaborating as closely as we should have been (especially given how our dashboards are meant to work in tandem). This was due mainly to us misunderstanding our objectives but we have corrected that this week and will continue to work closely moving forward. Additionally, I want to make sure that my work is accurately reflecting the objectives Professor Krishna has outlined for me. Moving forward, I hope to have more mid-week checkins with him so we are both on the same page.

11. Next Steps

- Continue to drill down if there are trust issues for the four issues seeing call volume decreases by examining where callers are disconnecting for each menu prompt, implying callers may be hanging up early due to discomfort with sensitive topics, pointing to trust, stigma, or usability barriers
- Work closely with teammates analyzing other metrics to link time-trend findings with their insights for a more complete understanding

12. References

- Input code from Krish
- Julia’s data transformations in python
- Information from Cynthia during class with her

Presentation 2

13. Link to presentation and code on GitHub:
    - Presentation Link: Click Here (Note: static visuals used in presentation; to see live dashboard use link below)
    - Dashboard Link: Click Here
    - Data cleaning code: see Julia's code attached below
        - Click Here
        - Click Here

14. What did you do?

This week, my goal was to build off my previous visualization—which tracks call volumes by month—by allowing the viewer to see call volumes for each menu while adding hierarchical filtering by year and month. This way, Cynthia can easily compare how many callers are there for each menu across the relevant time frames that she needs. This visual is supposed to compliment Julia's visual as they seek to display similar information across different time frames; Julia displays each menu's usage over the course of a day whereas mine shows it over the course of a year.

To do so, I had to first download the new CAR data for the month of September. Then, using Krish's input code and Julia's data cleaning and hierarchical filtering code, I created three .csv files. One was the combined the raw data into a single .csv (using Krish's code); another had the data grouped by 'Contact Session ID' while also calculating and creating columns to track the month and year of the call; and the last create a hierarchical structure across five tiers so each activity can be traced back through its broader categories.

Next, I uploaded the three data sets to Power BI where I linked the first dataset to the second by 'Contact Session ID' and linked the third dataset to the first by 'Activity Name.' Here, I made a clustered bar graph to display the number of unique call session ID's associated with a menu option on a monthly basis. With this in mind, the x-axis was the months in the year whereas the y-axis was the number of distinct 'Call Session IDs.' The graph does display a drop off in the number of calls Legal Aid received so far in 2025 compared to 2024. However, my hypothesis presented in class—that certain menu options that are politically charged right now (e.g. immigration, HIV, etc.) may be driving the drop off—proved to be wrong. In fact, there appeared to be a relatively proportional drop off across all menu options, which makes me wonder why there just appears to be fewer people calling in. I hope to investigate this further in the coming weeks.

15. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it currently looks like call volumes on track to be less for 2025 compared to 2024. Additionally, it appears that call volumes across all menu options reduced proportionally between 2024 and 2025. We need to understand why this is occurring before making any concrete recommendations.

16. Issues faced, attempt to resolve issues, and resolved issues:

I didn’t face many significant issues during this revision. In the past weeks, I think Julia and I were not collaborating as closely as we should have been (especially given how our dashboards are meant to work in tandem). This was due mainly to us misunderstanding our objectives but we have corrected that this week and will continue to work closely moving forward.

17. Next Steps

- Continue to investigate why call volumes dropped off between 2025 and 2024 across all menu types. I will dig through the data to see if there are other trends worth analysing to help answer this question.
- Work closely with teammates analyzing other metrics to link time-trend findings with their insights for a more complete understanding

18. References

- Input code from Krish
- Julia’s data transformations in python
- Information from Cynthia during class with her

Presentation 1

1. Link to presentation and code on GitHub:
    - Presentation Link: Click Here (Note: if the Power BI embedded in the slides fails, static visuals can be found at the bottom of the slide deck)
    - Coding was done on PowerBI. The data was originally transformed by my partner, Julia Schaffner, and then I built the visualization per the process below.

2. What did you do?

I created a visual for the Time Trends Dashboard using the CAR data. I built off my group partner, Julia's, work in which she compiled all CAR data and All Calls data into singular CSV files using the import code provided by Prof. Krishna. She then loaded them into PowerBI to perform more specific data transformations. Within Power BI, she used the Group By and Advanced Group By functions to ensure each Contact Session ID in the CAR dataset appeared only once, preventing double counting from multiple actions per session. Julia then created four new aggregated columns—Count, Call Start Time, Starting Hour, and All Rows—and expanded the nested tables to access detailed session information in the visuals.

With this data, I created a line chart visualization that aimed to examine if there was a seasonal peak in the number of calls received by Legal Aid. I made sure to filter the data so it only displays call volumes from 8/1/2024 - 7/31/2025 to ensure no double counting. The X-axis is the month in which the call took place over the course of this calendar year. The Y-axis is the number of unique calls during that month. On each visual, you can filter by EP Name, Flow Name, Termination Reason, and the axes for the visual.

3. How does it help the project?

Understanding when call volumes peak is crucial for improving the efficiency of the call system, especially given the constraints legal aid organizations face. With limited budgets and unpredictable funding due to shifting policies and economic conditions, Legal Aid is already struggling to meet demand. Calls often exceed capacity, leaving people unable to get through or facing long wait times. Since demand likely fluctuates rather than remaining constant—spiking during particular days, weeks, or policy changes—identifying these seasonal or situational trends allows the organization to anticipate and prepare for them. Though more analysis is needed, it currently looks like call volumes acutely spike in the new year, dip moderately during the summer months, and drop off dramatically in December. With this in mind, Legal Aid could leverage short-term staff or AI-powered agents during the new year while cutting back during the summer and the holidays. By aligning staffing and resources with periods of highest demand, the organization can serve more clients effectively without increasing costs, improving both accessibility and overall system performance.

4. Issues faced, attempt to resolve issues, and resolved issues:

The major issue I, along with the rest of my group, are facing is that the data has a lot of confusing and/or missing terms and information. We were able to get some context about acronyms and terms used by searching the internet or referencing the document Professor Arvind uploaded to GitHub; however, we still have questions about how the different answers in a column compare to each other in context of the overall business problem (i.e. SIP vs WXCC vs. WXC).

5. Next Steps

   - Drill down further to see if the call volume trend observed above is uniform across all menu types or if specific menu options follow unique trends and thus require different solutions/recommendations

- Compare if the seasonal spike from April to September is true in other years suggesting a regular trend or if it is more closely related to the extreme policy changes from this summer; to do so, I would create this same visualization with data from years past
- Refine the dashboard visuals to clearly display trends in peak call times by current filters; to do so, I would create heatmaps or bar graphs to highlight patterns across hours and days
- Add more filtering options to allow deeper analysis of when and why specific termination reasons happen
- Work closely with teammates analyzing other metrics to link time-trend findings with their insights for a more complete understanding

6. References

- Input code from Krish
- Julia's data transformations in python and PowerBI
- Information from Cynthia during class with her